PSEUDOCODES FOR BLASTSMART APP

1.Start

2.Print Title

- DISPLAY('Surface Blasting Design Calculator')

3. Use INSERTS variable values for equation

- BD =Input blasting diameter
- P =Input average in hole density
- Input Spacing to burden ratio
- Input bench height.

3. add additional fixed variables

    Powder factor = 0.3 #standard powder for limestone

4.CALCULATION

If user inputs yes for calculation to calculate

    Stemming_length= 25*BD/1000

    OUTPUT = Stemming_length

    MC= p*BD*BD/1273

    CL=BH-Stemming_length

    Burden= ((MC*CL)/(s_b_ratio*BH*PFtech))**(1/2)

    OUTPUT = Burden

    S= s_b_ratio*Burden

    H = CL + Stemming_length

    actual_pf= ((BH-Stemming_length)*MC)/(Burden*S*H)

    OUTPUT = actual_pf

Else if user inputs no

    OUTPUT = no calculations will be performed

ELSE

OUTPUT = USER PUT IN INVALID SYNTAX

PSEUDOCODE 2 FRAGMENTATION

1. Input Rock factor (A)
2. Input Average in-hole density (p)
3. Input Bench height (BH)
4. Input Relative effective energy of explosives(E)
5. Input Stemming length(SL)
6. If Explosive = ANFO

   E = 100

   elif Explosive = TNT

   E = 115

   else

   E = 150
7. Set Q = BH – SL
8. Set Mean fragment size = A * (K**(-0.8)) * (Q**(1/6)) * ((115/E)**(19/20))
9. Print Q
10. Print Mean fragment size